

COUNTING PROCESSES WITHIN MARKOV RENEWAL PROCESSES

Y.P. Gupta: National Institute of Health and Family Welfare,
New Mehrauli Road, New Delhi-110067 (India).

and
Sunita Gupta: Department of Mathematical Statistics, Lady Shri
Ram College for Women, University of Delhi, Lajpat
Nagar, New Delhi (India).

Key words:

Summary: In a Markov Renewal Process with $(m+1)$ states $0, 1, 2, \dots, m$, a counting process could be formulated by counting the number of visits to the states $1, 2, \dots, m$, between the successive visits to state 0. Here we derive results for the probability generating function and moments of the counting process within a Markov Renewal Process (M.R.P.), General Markov Renewal Process (G.M.R.P.), Equilibrium Markov Renewal Process (E.M.R.P.) and Cumulative Process. The results obtained by Gupta and Sharma (1976) for the Poisson counts of an M.R.P., G.M.R.P., E.M.R.P. and cumulative process associated with M.R.P. have been derived as particular cases.

1. INTRODUCTION

Pyke (1961a) defined a Markov Renewal Process with m (say) states as one which records at each time t , the number of times a system has visited each of the m states in the interval $(0, t)$, if the system moves from one state to another according to a Markov chain with transition probability matrix

$P_0 = (p_{ij})$ and the time required for each successive move is a random variable whose distribution function (d.f.) depends upon the two states between which the move is being made. Now, let us consider a Markov Renewal Process with $(m+1)$ states $0, 1, 2, \dots, m$ ($m < \infty$). We form a counting process by counting the number of visits to the states $1, 2, \dots, m$ between the successive visits to the state 0 . For example, in the usual gear box problem, when a gear box fails, it is replaced by another and the failed one is repaired and used again. If it fails once again, it is repaired and used. Since each repair affects the life distribution, the repaired gear boxes will be different among themselves depending upon the number of times they have been repaired and also from the new ones. If we assume that a gear box can be repaired m times at most and denote the new one by state 0 , repaired once by state 1 and so on, we shall have $(m+1)$ types of gear boxes available.

In this paper, the results for the counting process within a Markov Renewal Process, General Markov Renewal Process, Equilibrium Markov Renewal Process and cumulative process associated with an M.R.P. have been derived. It has also been shown that the Poisson counts of an M.R.P., G.M.R.P., E.M.R.P. and cumulative process associated with an M.R.P. obtained by Gupta and Sharma (1976), follow as a particular case of the results obtained in this paper.

2. NOTATIONS AND PREVIOUS RESULTS

Throughout this paper, following Pyke (1961a,b), we shall denote by

$$(i) \quad Q_{ij}(x) = P[J_n = j; \quad t_n - t_{n-1} \leq x | J_0, J_1, \dots, J_{n-1} = i, \\ t_1, t_2, \dots, t_{n-1}]$$

where $[J_n; n \geq 0]$ is a Markov chain with finite state space $(1, 2, \dots, m)$ and t_n denotes the n -th transition epoch.

$$\text{Also } H_i(t) = \sum_{j=1}^m Q_{ij}(t).$$

(ii) $F_{ij}(0) = \text{d.f. of sojourn time in state } i \text{ before making a transition to state } j.$

(iii) $N(t) \equiv \sup(n \geq 0, t_n \leq t)$ denotes the total number of transitions the system undergoes during the interval $(0, t)$.

Therefore $N(t) = \sum_{j=1}^m N_j(t)$ where $N_j(t)$ denotes the number of transitions to state j up to time t .

(iv) $Z_t = J_{N(t)}$ denotes the state of the system at time t , after the $N(t)$ -th transition.

The Laplace-Stieltjes transforms (L-S.T.) will be denoted by lower case letters, e.g. L-S.T. of $Q_{ij}(t)$ by $q_{ij}(s)$ etc.

(v) $G_{ij}(t) = P[N_j(t) > 0 | Z_0 = i]$

(vi) $W_{ij}(r, t) = \text{p.g.f. of } P[N_j(t) = n_j | Z_0 = i],$
 $r = (r_{ij}), |r_{ij}| \leq 1, i = 1, 2, \dots, m; j = 1, 2, \dots, m.$

(vii) $w_{ij}(r, s) = \text{L-S.T. w.r.t. } t \text{ of } W_{ij}(r, t).$

(viii) $N_j(t_0, t) = \text{The number of times } j\text{-th state has been visited by the system in an arbitrary interval } (t_0, t_0 + t), t_0 \neq 0.$

(ix) $w_{ij}(r, t_0, s) = \text{L-S.T. w.r.t. } t \text{ of the p.g.f. of } P[N_j(t_0, t) = n_j | Z_0 = i]$

(x) $w_{ij}^*(r, s_0, s) = \text{Laplace Transform w.r.t. } t_0 \text{ of } w_{ij}(r, t_0, s).$

The expressions in matrix form of $w_{ij}(r,s)$ and $w_{ij}^*(r,s_0,s)$ have been obtained by Kshirsagar and Wysocki (1970) as given below:

$$w^*(r,s_0,s) = \frac{1}{s_0} E + \frac{1}{s-s_0} [(I-q(s_0))^{-1} [(q(s_0)-q(s)) \square r] w(r,s)] - (q(s_0) - q(s)) I$$

$$\text{and } w(r,s) = (I-q(s) \square r)^{-1} (I-h(s))$$

where $h(s) = (\delta_{ij} h_i(s))$; $i = 1, 2, \dots, m$, $j = 1, 2, \dots, m$

$$E = (\underline{ee}'), \quad \underline{e}' = (1, 1, \dots, 1), \quad q(s) = (q_{ij}(s))$$

and for square matrices $A = (a_{ij})$ and $B = (b_{ij})$,

$$A \square B = (a_{ij} b_{ij}).$$

3. COUNTING PROCESS WITHIN AN M.R.P.

For a Markov Renewal Process with $(m+1)$ states, we denote the joint distribution of the number of transitions to various states in an arbitrary interval (t_0, t_0+t) by

$$p_{n_0, n_1, \dots, n_m}^i(t_0, t) = P[N_j(t_0, t) = n_j; j = 0, 1, \dots, m | Z_0 = i]$$

and

$$p_{n_1, n_2, \dots, n_m}^i(t_0, t) = P[N_j(t_0, t) = n_j; j = 1, 2, \dots, m | Z_0 = i]$$

The probability generating functions of $p_{n_0, n_1, \dots, n_m}^i(t_0, t)$ and

$p_{n_1, n_2, \dots, n_m}^i(t_0, t)$ are denoted respectively by

$$W_i(r^0, t_0, t) = \sum_{n_0, n_1, \dots, n_m=0}^{\infty} r_0^{n_0} r_1^{n_1} \dots r_m^{n_m} p_{n_0, n_1, \dots, n_m}^i(t_0, t)$$

and

$$(i) \quad W_i(r, t_0, t) = \sum_{n_1, n_2, \dots, n_m=0}^{\infty} r_1^{n_1} r_2^{n_2} \dots r_m^{n_m} p_{n_1, n_2, \dots, n_m}^i(t_0, t) \\ = W_i(r^0, t_0, t) |_{r_0=1}$$

where $r^0 = \text{diag}(r_0, r_1, \dots, r_m)$.

Let the system have visited the state 0 at the time points $T_0, T_1, \dots$. Then successive visits to state 0, starting from state 0 form an ordinary renewal process (called a zero process). The distribution function of the interval between successive visits to state 0 is given by $G_{00}(x) = R_0(x)$ (say).

Let $X_k = T_k - T_{k-1}$; $k = 1, 2, \dots$.

Then T_{k-1} and X_k are independently distributed. Let S_k denote the epoch of the k -th renewal. The p.d.f. $p(T_{k-1})$ of T_{k-1} is given by (Cox(1962))

$$p(T_{k-1})dT_{k-1} = [R_0^{k-2}(T_{k-1}) - R_0^{k-1}(T_{k-1})]h(T_{k-1})dT_{k-1}$$

where $R_0^k(x)$, the cumulative distribution function of S_k , is the k -fold convolution of the distribution function $R_0(x)$ with $R_0^0(x) = 1$ and the renewal density $h(T_{k-1})$ is

$$h(T_{k-1}) = \lim_{\delta T_{k-1} \rightarrow 0} \frac{E[N(T_{k-1}, T_{k-1} + \delta T_{k-1})]}{\delta T_{k-1}}$$

Define

$$p_{ik}(n_1, n_2, \dots, n_m) = P[N_j(T_{k-1}, T_k) = n_j, j = 1, 2, \dots, m | Z_0 = i] \\ i = 0, 1, 2, \dots, m, k = 1, 2, \dots,$$

Using the results given by Kshirsagar (1970), the above expression can be written as

$$(2) \quad \begin{cases} p_{ik}(n_1, n_2, \dots, n_m) = \int_0^\infty \int_0^\infty p_{n_1, n_2, \dots, n_m}^i(T_{k-1}, x_k) p(T_{k-1}) dR_0(x_k) dT_{k-1} \\ \quad \text{for } k \geq 2 \\ p_{i1}(n_1, n_2, \dots, n_m) = \int_0^\infty p_{n_1, n_2, \dots, n_m}^i(0, x_1) dR_0(x_1) \end{cases}$$

The generating function of $p_{ik}(n_1, n_2, \dots, n_m)$ is given by

$$G_i(\alpha, r) = \sum_{k=1}^{\infty} \alpha^k \sum_{n_1, n_2, \dots, n_m=0}^{\infty} r_1^{n_1} r_2^{n_2} \dots r_m^{n_m} (p_{ik}(n_1, n_2, \dots, n_m))$$

Using (1) and (2), we have

$$(3) \quad G_i(\alpha, r) = \alpha \int_0^\infty W_i(r, 0, x) dR_0(x) + \sum_{k=2}^{\infty} \alpha^k \int_0^\infty \int_0^\infty W_i(r, T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x)$$

To obtain the generating function of the counts within an M.R.P., put $i=0$ in (3). We get

$$(4) \quad G_0(\alpha, r) = \alpha \int_0^\infty W_0(r, 0, x) dR_0(x) + \sum_{k=2}^{\infty} \alpha^k \int_0^\infty \int_0^\infty W_0(r, T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x)$$

Now to obtain the moments, we have to differentiate $G_0(\alpha, r)$ w.r.t. r and put $r=1$

$$\begin{aligned} \frac{\partial G_0(\alpha, r)}{\partial r_j} &= \alpha \int_0^\infty \frac{\partial W_0(r, 0, x)}{\partial r_j} dR_0(x) \\ &+ \sum_{k=2}^\infty \alpha^k \int_0^\infty \int_0^\infty \frac{\partial W_0(r, T_{k-1}, x)}{\partial r_j} p(T_{k-1}) dT_{k-1} dR_0(x) \\ &\text{for } j = 1, 2, \dots, m \end{aligned}$$

$$\begin{aligned} \bar{M}_{0j}(\alpha) &= \left. \frac{\partial G_0(\alpha, r)}{\partial r_j} \right|_{r=1} \\ &= \alpha \int_0^\infty M_{0j}(x) dR_0(x) + \sum_{k=2}^\infty \alpha^k \int_0^\infty \int_0^\infty M_{0j}(T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0'(x) \\ &\text{for } j = 1, 2, \dots, m \end{aligned}$$

$$\text{where } M_{0j}(T_{k-1}, x) = \left. \frac{\partial W_0(r, T_{k-1}, x)}{\partial r_j} \right|_{r=1} \quad \text{and } M_{0j}(x) = M_{0j}(0, x)$$

Similarly

$$\begin{aligned} \bar{R}_{0j}(\alpha) &= \left. \frac{\partial^2 G_0(\alpha, r)}{\partial r_j^2} \right|_{r=1} \quad \text{for } j = 1, 2, \dots, m \\ &= \alpha \int_0^\infty R_{0j}(x) dR_0(x) + \sum_{k=2}^\infty \alpha^k \int_0^\infty \int_0^\infty R_{0j}(T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x) \\ &\text{for } j = 1, 2, \dots, m, \end{aligned}$$

where

$$R_{0j}(T_{k-1}, x) = \left. \frac{\partial^2 W_0(r, T_{k-1}, x)}{\partial r_j^2} \right|_{r=1}$$

$$\text{and } R_{0j}(x) = R_{0j}(0, x).$$

Also

$$\begin{aligned} C_{jk}^0(\alpha) &= \frac{\partial^2 G_0(\alpha, r)}{\partial r_j \partial r_k} \Big|_{r=1} \\ &= \alpha \int_0^\infty C_{jk}^0(x) dR_0(x) + \sum_{k=2}^\infty \alpha^k \int_0^\infty \int_0^\infty C_{jk}^0(T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x) \end{aligned}$$

for $j \neq k$

where

$$C_{jk}^0(T_{k-1}, x) = \frac{\partial^2 W_0(r, T_{k-1}, x)}{\partial r_j \partial r_k} \Big|_{r=1} \quad \text{and} \quad C_{jk}^0(x) = C_{jk}^0(0, x)$$

In the Markov Renewal Process with $(m+1)$ states, if we start from state i but not from the state 0 , we get a modified counting process (i.e. counting process within a G.M.R.P.). In this case for the first visit to state 0 , the distribution function will be $G_{i0}(x)$ and we get

$$\begin{aligned} (5) \quad \overline{G_i(\alpha, r)} &= \alpha \int_0^\infty W_i(r, 0, x) dG_{i0}(x) \\ &\quad + \sum_{k=2}^\infty \alpha^k \int_0^\infty \int_0^\infty W_i(r, T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x) \end{aligned}$$

Differentiating $\overline{G_i(\alpha, r)}$ w.r.t. r and putting $r=1$, we obtain the moments as shown below:

$$\begin{aligned} \frac{\partial \overline{G_i(\alpha, r)}}{\partial r_j} &= \alpha \int_0^\infty \frac{\partial W_i(r, 0, x)}{\partial r_j} dG_{i0}(x) \\ &\quad + \sum_{k=2}^\infty \alpha^k \int_0^\infty \int_0^\infty \frac{\partial W_i(r, T_{k-1}, x)}{\partial r_j} p(T_{k-1}) dT_{k-1} dR_0(x) \end{aligned}$$

$$\overline{M_{ij}(\alpha)} = \frac{\partial \overline{G_i(\alpha, r)}}{\partial r_j} \Big|_{r=1}$$

$$(6) \quad = \alpha \int_0^{\infty} M_{ij}(x) dG_{i0}(x) + \sum_{k=2}^{\infty} \alpha^k \int_0^{\infty} \int_0^{\infty} M_{ij}(T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x) \\ \text{for } j = 1, 2, \dots, m$$

Similarly we obtain

$$(7) \quad \overline{R_{ij}}(\alpha) = \frac{\partial^2 G_i(\alpha, r)}{\partial r_j^2} \Big|_{r=I} \\ = \alpha \int_0^{\infty} R_{ij}(x) dG_{i0}(x) + \sum_{k=2}^{\infty} \alpha^k \int_0^{\infty} \int_0^{\infty} R_{ij}(T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x) \\ \text{for } j = 1, 2, \dots, m$$

and

$$(8) \quad \overline{C_{jk}^i}(\alpha) = \frac{\partial^2 G_i(\alpha, r)}{\partial r_j \partial r_k} \Big|_{r=I} \\ = \alpha \int_0^{\infty} C_{jk}^i(x) dG_{i0}(x) + \sum_{k=2}^{\infty} \alpha^k \int_0^{\infty} \int_0^{\infty} C_{jk}^i(T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x) \\ \text{for } j, k = 1, 2, \dots, m; j \neq k.$$

4. COUNTING PROCESS WITHIN AN EQUILIBRIUM MARKOV RENEWAL PROCESS

Pyke (1961b) has proved that an E.M.R.P. is nothing but a G.M.R.P. with initial probabilities:

$$a_i = \frac{\eta_i}{\mu_{ii}}; \quad i = 1, 2, \dots, m$$

$$\text{and } \overline{Q_{ij}}(t) = \frac{p_{ij}}{\eta_i} \int_0^t [1 - F_{ij}(y)] dy, \quad i = 1, 2, \dots, m$$

where $\overline{Q_{ij}}(t) = P[J_1 = j, t_1 \leq t | J_0 = i]$, $i = 1, 2, \dots, m$, μ_{ii} is the mean of $G_{ii}(x)$ and $\eta_i = E[\sum_{j=1}^m Q_{ij}(t)]$.

Kshirsagar and Gupta (1969) have shown that the L-S.T. of the absolute means of $[N_1(t), N_2(t), \dots, N_m(t)]$ is given by

$$\frac{1}{s} \left[\frac{1}{\mu_{11}}, \frac{1}{\mu_{22}}, \dots, \frac{1}{\mu_{mm}} \right].$$

Since $M_{ij}(t_0, x) = M_{ij}(t_0 + x) - M_{ij}(t_0)$, it follows from (3.6) that

$$\begin{aligned} \overline{M_{ij}}(\alpha) &= \alpha \int_0^{\infty} M_{ij}(x) dG_{i0}(x) \\ &\quad + \sum_{k=2}^{\infty} \alpha^k \int_0^{\infty} \int_0^{\infty} [M_{ij}(T_{k-1} + x) - M_{ij}(T_{k-1})] p(T_{k-1}) dT_{k-1} dR_0(x). \end{aligned}$$

If $(a_0, a_1, \dots, a_m)$ be the vector of initial state probabilities, the absolute moments are given by

$$\sum_{i=0}^m a_i \overline{M_{ij}}(\alpha) = \frac{\alpha}{\mu_{jj}}$$

Therefore the absolute moments in the case of a counting process for an E.M.R.P. are given by

$$\begin{aligned} \sum_{i=0}^m a_i \overline{M_{ij}}(\alpha) &= \alpha \int_0^{\infty} \sum_{i=0}^m a_i M_{ij}(x) dG_{i0}(x) \\ &\quad + \sum_{k=2}^{\infty} \alpha^k \int_0^{\infty} \int_0^{\infty} \frac{x}{\mu_{jj}} p(T_{k-1}) dT_{k-1} dR_0(x) \\ &= \alpha \int_0^{\infty} \sum_{i=0}^m a_i M_{ij}(x) dG_{i0}(x) + \sum_{k=2}^{\infty} \alpha^k \frac{\mu_{00}}{\mu_{jj}}. \end{aligned}$$

Thus except for $k=1$, the absolute mean number of visits to state 0 is given by $\frac{\mu_{00}}{\mu_{jj}}$ which is quite logical.

5. COUNTING PROCESS WITHIN A CUMULATIVE PROCESS

Kshirsagar and Gupta (1969) have defined a cumulative process U_t associated with an M.R.P. with m states as

$$U_t = \sum_{r=1}^{N(t)} W_r, N(t) = 1, 2, \dots$$

Further, we assume that the d.f. of W_r is $Y_j(w)$ if $J_r = j$ ($j = 1, 2, \dots, m$). Then the counts within a cumulative process can be defined as

$U_k = \sum W_r$ where the summation is over all the transitions in the interval (T_{k-1}, T_k) . Let

$$N_j(T_{k-1}, T_k) = n_j \quad (j = 1, 2, \dots, m).$$

Then the L-S.T. of the d.f. of U_k , conditional on $Z_0 = 0$ is $y_1^{n_1}(p) y_2^{n_2}(p) \dots y_m^{n_m}(p)$ if $n_1, n_2, \dots, n_m$ are fixed and $y_j(p)$ is the L-S.T. of $Y_j(w)$, the d.f. of W_r . But the n_j ($j=1, 2, \dots, m$) are also random variables and hence the unconditional L-S.T. of the d.f. of U_k is given by

$$\begin{aligned} & \sum_{n_1, n_2, \dots, n_m=0}^{\infty} \prod_{j=1}^m y_j^{n_j}(p) P\{N_j(T_{k-1}, T_k) = n_j, \quad j=1, 2, \dots, m | Z_0 = 0\} \\ &= \sum_{n_1, n_2, \dots, n_m=0}^{\infty} \prod_{j=1}^m y_j^{n_j}(p) p_{0k}(n_1, n_2, \dots, n_m). \end{aligned}$$

Hence the generating function of counts of a cumulative process is given by

$$G_0(\alpha, y(p)) = \sum_{k=1}^{\infty} \alpha^k \sum_{n_1, n_2, \dots, n_m=0}^{\infty} \prod_{j=1}^m y_j^{n_j}(p) p_{0k}(n_1, n_2, \dots, n_m)$$

where $y(p) = \text{diag}[y_1(p), y_2(p), \dots, y_m(p)]$

Using (3.1) and (3.2), we get

$$(9) \quad G_0(\alpha, y(p)) = \alpha \int_0^\infty w_0(y(p), 0, x) dR_0(x) \\ + \sum_{k=2}^{\infty} \alpha^k \int_0^\infty \int_0^\infty w_0(y(p), T_{k-1}, x) p(T_{k-1}) dT_{k-1} dR_0(x)$$

The moments of a counting process within a cumulative process can be obtained by differentiating (9) and putting $y(p) = E$ similar to that done in earlier sections.

6. PARTICULAR CASES

If we put $dR_0(x) = \gamma e^{-\gamma x} dx$ i.e. the d.f. of first passage time to state 0 is exponential, then

$$p(T_{k-1}) = \frac{\gamma(\gamma T_{k-1})^{k-2} e^{-\gamma T_{k-1}}}{(k-2)!}$$

and equation (3.4) after simplification leads to

$$G_0(\alpha, r) = \alpha w_0(r, \gamma) + \alpha^2 \gamma w_0^*(r, \gamma(1-\alpha), \gamma)$$

$$\text{where } w_0^*(r, \gamma(1-\alpha), \gamma) = \int_0^\infty w_0(r, T_{k-1}, \gamma) e^{-\gamma T_{k-1}(1-\alpha)} dT_{k-1}$$

or in matrix notation,

$$G(\alpha, r) = \alpha w(r, \gamma) + \alpha^2 \gamma w^*(r, \gamma(1-\alpha), \gamma)$$

which is the same as obtained by Gupta and Sharma (1976).

Similarly we can show that the results for the counting distributions for G.M.R.P., E.M.R.P. and cumulative process obtained by Gupta and Sharma (1976) follow as particular cases.

ACKNOWLEDGEMENT

The authors are thankful to Professor Kanwar Sen, Head, Department of Mathematical Statistics, University of Delhi, for valuable suggestions. Thanks are also due to the Referee for making valuable suggestions.

REFERENCES

- COX, D.R. (1962). *Renewal Theory*. Methuen and Co. Ltd., London.
- GUPTA, Y.P. and SHARMA, S.C. (1976). Some results in Markov renewal processes, *Metron*, Vol. **34**, No. 1-2, 81-92.
- KSHIRSAGAR, A.M. (1970). Poisson counts of a Markov Renewal process. *South African Statist. J.*, **4**, 67-72, ST: 12-65, Zb: 223-578.
- KSHIRSAGAR, A.M. and GUPTA, Y.P. (1969). Some results in Markov renewal processes. *Cal. Stat. Assoc. Bull.*, **18**, 61-72.
- KSHIRSAGAR, A.M. and WYSOCKI, R. (1970). Some distribution and moment formulae for the Markov Renewal process. *Proc. Camb. Phil. Soc.*, **68**, 159-166.
- PYKE, R. (1961a). Markov renewal processes: definitions and preliminary properties. *Ann. Math. Stat.*, **32**, 1231-1242.
- PYKE, R. (1961b). Markov renewal processes with finitely many states. *Ann. Math. Stat.*, **32**, 1243-1259.